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Mechanism of ultrafast laser welding defects in dissimilar materials of quartz glass to 304 stainless steel

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Abstract

Glass-to-metal welding is very vital for advanced devices in aerospace, sensors, micro-electro-mechanical systems and other fields, but it still remains challenging due to property mismatches that cause porosity and cracks which severely degrade joint integrity. While ultrafast laser welding is a promising technique, most research has focused on microstructures and mechanical properties of the joints, with a lack of systematic studies on the formation mechanisms of defects. This study investigated the picosecond laser welding of quartz glass to 304 stainless steel and explored the effects of the defocus distance and laser power on the formation of welding defects. The evolution and mechanisms of typical welding defects were revealed through comprehensive characterization using SEM, EDS, XPS, and TEM. The results show that the defocus distance and laser power dictate a transition in the dominant energy deposition mechanism from "glass-volume-dominated" to "metal-surface-dominated". In the latter regime, intense metal vaporization creates the confined high-pressure vapor that penetrates the molten glass, forming porosity, with the vapor also inducing micro-cracks and incomplete bonding. Based on these, an "energy deposition mechanism competition" model was proposed to provide a theoretical basis for understanding defect formation and guiding process optimization, which successfully suppressed defects and achieved a five-fold increase in the average joint shear strength, reaching a maximum of 378.41 N. This study can demonstrate a viable pathway for high-quality glass-to-metal welding via controlling energy distribution and greatly broaden the application of heterogeneous joints in manufacturing industry.

Keywords

Picosecond laser welding; Dissimilar materials of glass and metal; Welding defects; Forming mechanism; Process control

1. Introduction

In recent years, the dissimilar materials welding of glasses to metals has received widespread attention because it combines the excellent workability and ductility of metal materials with the superior optical properties and corrosion resistance of glass, which can be widely applied in optical sensors, medical implants, microfluidic devices, MEMS devices, microelectronic devices and other field [[1], [2], [3], [4]]. Nevertheless, the significant discrepancy in physical and chemical properties between glasses and metals easily leads to the formation of defects such as porosity [5] and cracks [6,7] during the welding process, which make it very difficult to find a practical, defect-free and cost-effective technique for precise and robust bonding between them. Therefore, how to suppress welding defects and improve joint quality remains a key challenge for the joining of glass-metal dissimilar materials.

In order to obtain a joint with good weld quality, various welding methods, involving electron beam welding [8], ultrasonic welding and ultrasonic-assisted welding [9,10], soldering [11,12], diffusion bonding [13], laser welding [14,15] and so on, were attempted for joining glass to the same or dissimilar materials. Although these methods can achieve the bonding of glasses and metals, it is easy to produce cracks at the joint interface. This can be attributed to thermal stress induced by differences in thermal expansion coefficients [16,17], or to the formation of brittle interfacial products, such as oxide phases or reaction layers, that readily initiate cracks under stress [18,19]. The occurrence of cracks can reduce the mechanical properties and air tightness of the glass-metal joints, which greatly restricts the application of heterogeneous components in the industrial field.

Compared with the traditional methods, nanosecond laser welding has the advantages of high precision, low heat-affected zone, small thermal deformation and suitability for welding dissimilar materials [[20], [21], [22]]. Some efforts have been made by researchers to connect dissimilar materials of glasses and metals. Huo et al. [23,24] systematically investigated the welding of silica glass to 304 stainless steel using an infrared nanosecond fiber laser. Their work revealed the preferential diffusion of Cr over Fe into the glass during welding and demonstrated that introducing a Cr interlayer could enhance joint strength by reducing microcracks caused by thermal expansion mismatch. In a different approach, A. de Pablos-Martín and Höche [25] conducted nanosecond pulsed laser welding of borosilicate glasses with titanium intermediate layer, and determined that the origin of the crack in the glass/Ti/glass samples was probably because of the thermal shock generated by the laser interaction with the sample. Although these strategies can enhance the shear strength of glass-metal joints, crack formation remains a persistent challenge during nanosecond laser welding. The occurrence of cracks in joints can reduce the service life of the joint and compromise on safety hazards. The development of new welding processes to suppress cracks and effectively improve joint quality has become the focus of current research.

Ultrafast laser refers to laser technology with pulse widths ranging from picoseconds (10^{-12} s) to femtoseconds (10^{-15} s) [26]. On account of the extremely short pulse duration and high peak power, ultrafast laser welding possesses the advantages of high accuracy and small thermal effect, which is regarded as an ideal method for connecting various glasses and metals [[27], [28], [29], [30]], and initial studies demonstrated its feasibility and revealed characteristic challenges. Wang et al. [31] investigated the effects of femtosecond laser parameters on the weld quality, and revealed that no new phases were generated between the glass and 304 stainless steel and that the joining mechanism was mainly composed of mechanical mixing between the two materials. Similarly, Yang et al. [32] conducted ultrafast laser welding of quartz glass and 304 stainless steel, and found that a large number of shrinkage cavities were formed in the glass. The occurrence of shrinkage cavities was mainly attributed to the lack of internal material resulting from the substantial quartz used to fill the gap between glass and steel. Studies on other glass-metal systems provide complementary insights into interfacial reactions and process dynamics. Ciuca et al. [33] studied the microstructure and interface reaction behaviour in glass-aluminium lap joints, and found that the microstructure in the weld zone consisted of nanocrystalline silicon and transitional alumina phases, and a reduction of the silica glass led to the formation of amorphous alumina. Zhan et al. [34] researched the transient interaction between aluminum and fused silica, and confirmed that it was hard to avoid the inner stress at high pulse energy due to strong non-linear absorption, and the joint shear strength can be improved by adopting appropriate pulse energy and focus pre-compensation during the femtosecond laser welding process. Alongside these investigations into fundamental mechanisms, parallel research efforts have been directed toward process optimization to enhance weld integrity. Li et al. [35] carried out an integrated investigation on femtosecond laser welding of glass and metal, and determined that the heat accumulation during the welding process can increase laser-induced thermal stress within the glass, thus resulting in poor weldability. Morawska et al. [36] performed picosecond laser welding of stainless steel to quartz, and demonstrated that acceptable weld performance can be achieved by carefully optimizing the processing parameters (pressure, cleaning, average power, defocus etc.). Notably, the stresses and complex thermomechanical interactions inherent to the welding process can lead to a wider spectrum of defects, including cracks and pores. Wang et al. [37] investigated femtosecond laser welding of glass to an aluminum film, and determined that the size of defects within the weld seam is critical to joint integrity. They found emergent gaps and low-density regions lead to extremely poor strength and robustness, whereas achieving a defect-free, seamlessly integrated weld zone with nanoscale imperfections is essential for obtaining high-strength connections. Chen et al. [38,39] investigated the regulation of the molten pool in picosecond laser welding of borosilicate glass, and identified that the longitudinal over-expansion of the laser-affected zone leads to microcracks. By improving the Gerchberg–Saxton algorithm to homogenize energy distribution, these defects were effectively suppressed.

At present, most of the research on ultrafast laser welding of glasses and metals focuses on microstructure and mechanical properties, while there are few reports on the formation mechanisms of key welding defects such as porosity and cracks. Defects are a key factor affecting the mechanical property and service life of heterogeneous joints. Therefore, revealing the forming mechanism of defects in the ultrafast laser welding process is of great significance for improving the formability and mechanical properties of the joints.

In this study, infrared picosecond laser was used for the connection of quartz glass and 304 stainless-steel. The effects of welding parameters on weld formation were carefully investigated. The evolution law and formation mechanism of typical welding defects have been deeply explored using scanning electron microscope (SEM), energy dispersive spectroscopy (EDS), X-ray photoelectron spectroscopy (XPS) and transmission electron microscope (TEM). On this basis, a physical model was proposed to reveal the formation mechanism of welding defects. Moreover, possible improvements to the welding process have been proposed to achieve a defect-free joint with good formability and mechanical property.

2. Experimental materials and methods

The materials used for experiments in this study were quartz glass with the dimensions of φ30 mm×1 mm and SUS304 stainless steel with the dimensions of φ30 mm×7 mm. The chemical compositions of experimental quartz glass and SUS304 stainless steel are presented in Table 1.

Table 1. The chemical composition of the base materials (Mass fraction. %).

Element	**C**	**Si**	**Mn**	**Cu**	**Ni**	**Cr**	**S**	**P**	**Fe**
SUS304	0.08	0.06	1.28	1.67	9.30	17.61	0.001	0.019	69.98
Element	Si	O							
quartz glass	44.74	55.26							

The schematic diagram of ultrafast laser welding device and laser scanning strategy is shown in Fig. 1. The experimental welding devices were mainly made up of infrared picosecond laser, focusing system, self-made fixture and control units. The laser adopted in the experiments is a 1030 nm ps pulsed solid state laser (Huaray, pine2-1030-60).

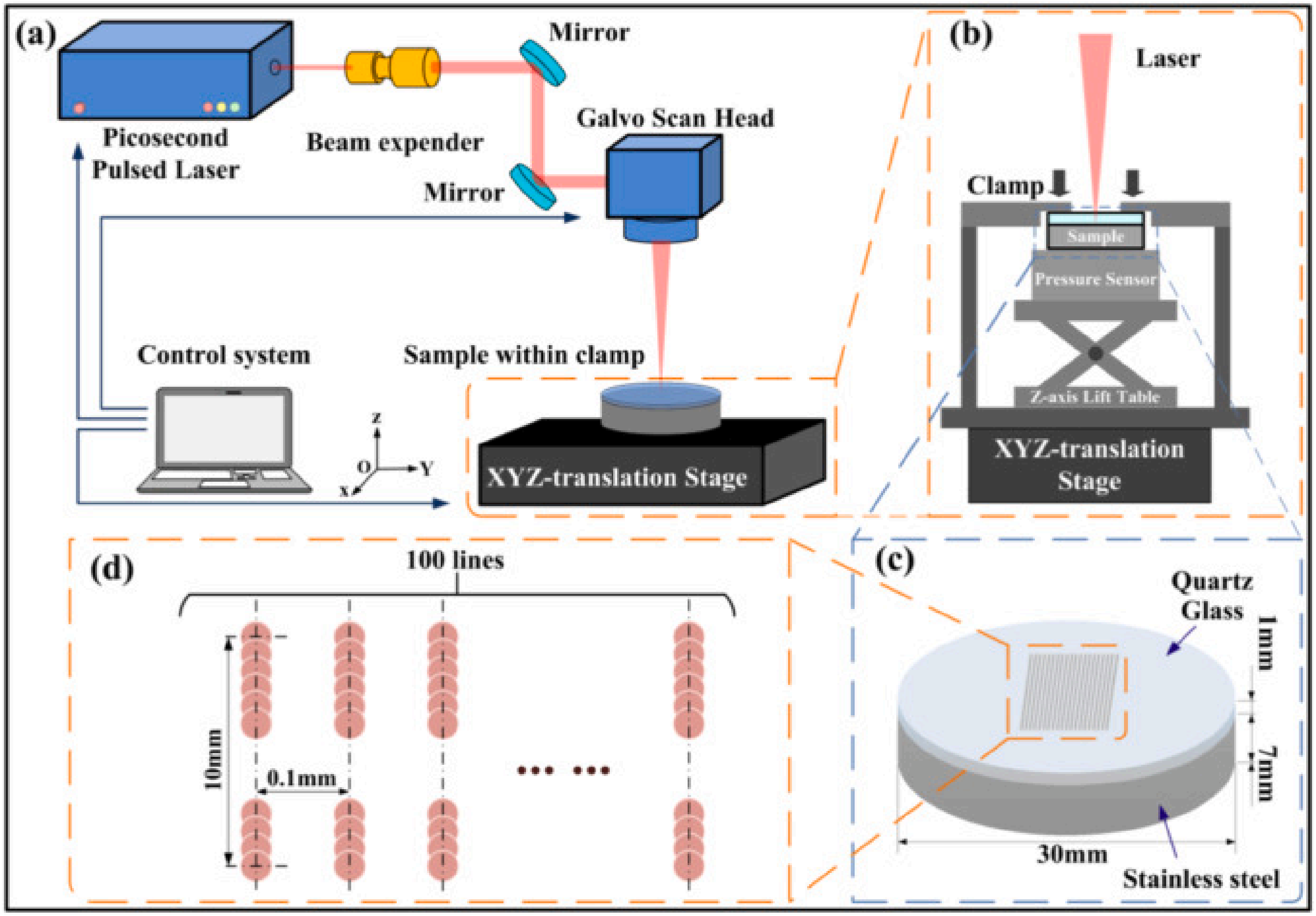

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Fig. 1. Schematic diagram of ultrafast laser welding: (a) welding device; (b) fixture device; (c) samples and (d) scanning strategy.

The main parameters of infrared picosecond laser are as follows: maximum laser power of 60W, pulse width of 10 ps, and repetition rate of 400-1000kHz. The focusing system in the welding device is a galvanometer scanner (Jenoptik, JENar) which deflects the laser beam at high speed through electromagnetic torque. The beam was then focused through an f-θ lens with a 170mm focal length. The sample assembly was mounted in a custom fixture, as shown in Fig. 1(b). The clamping force was applied by raising the Z-axis lift table, pressing the sample against the upper plate to generate a controlled pressure, which was monitored in real-time by the pressure sensor beneath the sample. The welding samples consisted of a 304 stainless steel disc and a quartz glass disc assembled without any intermediate interlayer. The steel disc was positioned at the bottom, with the glass disc on top. Their precise dimensions are detailed in Fig. 1(c). The contact surfaces were prepared following standard grinding and polishing procedures to achieve partial optical contact before welding. The final surface roughness was quantitatively measured using a stylus profilometer (Bruker, Dektak XT). The arithmetic mean roughness (Ra) was approximately 0.028 μm for the stainless-steel surface and 0.01 μm for the quartz glass surface. The welding zone and laser scanning path are depicted in Fig. 1(c) and (d). The laser beam was directed by f-θ lens system, scanning in a unidirectional pattern over a suitably central location on the sample. The process involved 100 parallel tracks with a line spacing of 0.1 mm, each 10 mm in length, defining the total processed area.

The main process parameters involved pulse laser power, welding speed, repetition frequency and defocus distance during picosecond laser welding. A negative defocus value indicates that the laser focus was positioned below the metal-glass interface, with the absolute value representing the specific distance between the interface from the focal plane. The specific process parameters are presented in Table 2.

Table 2. Process parameters of ultrafast laser welding.

Equipment	**Variables**	**Units**	**Levels**
Infrared picosecond laser	Pressing force	N	400
	Pulse laser power	W	8; 12; 16; 20
	Repetition frequency	kHz	1000
	Welding speed	mm/s	50
	Defocus distance	mm	−0.1; −0.3; −0.5; −0.7

During the laser welding process, the pressing force was also imposed on the surface of the workpieces to ensure that the contact surfaces of welded samples are fully combined. Following welding, the joints were characterized to evaluate their microstructure, composition, and mechanical properties. Metallographic specimens were first prepared for cross-sectional analysis. Due to the small weld seam and the non-conductive nature of the glass component, the samples were precisely sectioned using a diamond wire saw, cold-mounted in resin, and then ground and polished to obtain standard metallographic surfaces.

The prepared specimens were initially examined using an optical microscope to assess the overall weld morphology and formation under different processing parameters. For detailed microstructural and elemental analysis, the samples were sputter-coated with a thin Au layer and subsequently investigated by scanning electron microscopy (SEM; ZEISS GeminiSEM) operated at 15 kV. The backscattered electron (BSE) imaging mode was primarily used to observe microstructural features near the SUS304/quartz glass interface. Energy dispersive spectroscopy (EDS; Oxford Aztec) was further employed for elemental mapping to analyze the origin and evolution of typical defects, such as porosity and cracks, and to explore their formation mechanisms.

The mechanical performance of the welded joints was evaluated via shear testing using an electrodynamic dynamic and static fatigue testing machine (Instron E1000). Tests were conducted at a constant displacement rate of 0.3 mm/min until failure. For each set of welding parameters, the test was repeated three times, and the average value of the maximum load was used to quantify the joint strength. Based on the comprehensive correlation of process parameters, microstructural characteristics, and mechanical properties, potential improvements to the welding process were proposed to inhibit defects and achieve joints with sound formation and superior mechanical performance.

3. Results and discussion

3.1. Weld formation

In order to investigate the effects of process parameters on weld formation, comparative ultrafast laser welding tests were carried out on dissimilar materials of 304 stainless steel and quartz glass. The main welding process parameters in the experiments are as follows: 1000 kHz pulse frequency, 16 W laser power and 50 mm/s welding speed. The microstructure at the interface of the joints under different defocus conditions is shown in Fig. 2. Notably, cracks are observed across all defocus conditions, but their characteristics and the overall defect landscape evolve distinctly with defocus distance.

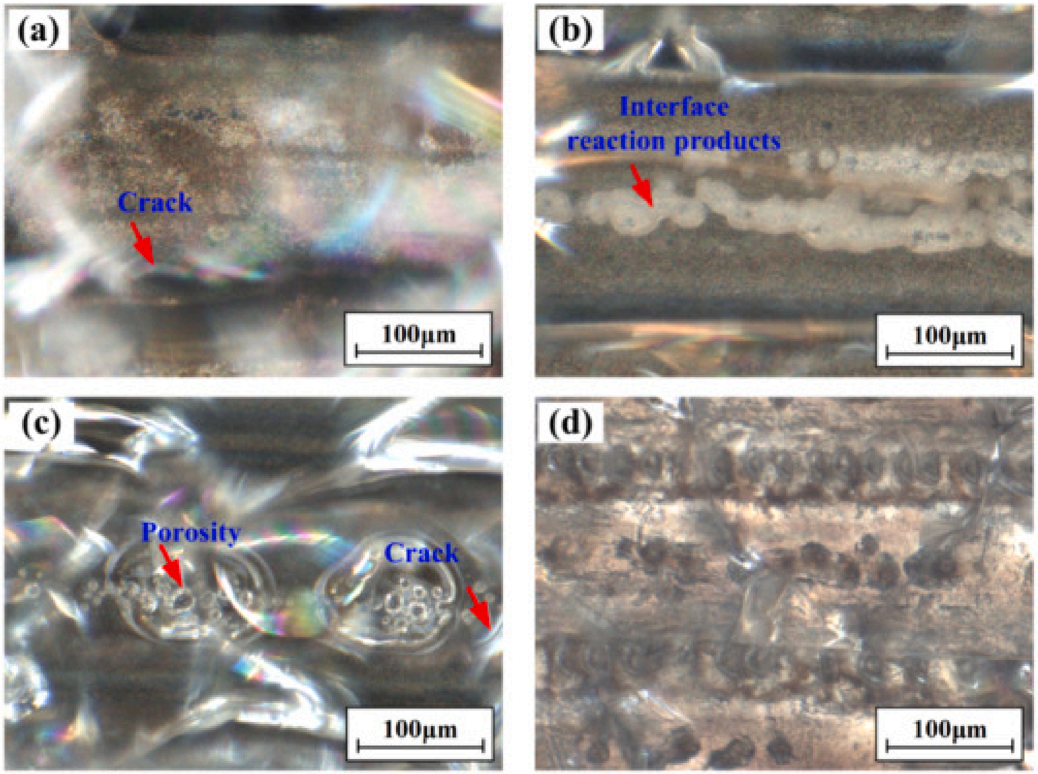

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Fig. 2. Macroscopic morphology of the joint under different defocus conditions: (a) −0.1 mm; (b) −0.3 mm; (c) −0.5 mm; (d) −0.7 mm.

As shown in Fig. 2, weld morphology varies significantly with defocus distance. At the −0.1 mm defocus condition in Fig. 2(a), colorful interference patterns appear on the glass surface. These patterns originate from the optical defocusing of subsurface micro-cracks that lie outside the focal plane of the microscope, indicating that the dominant defects are fine cracks distributed within the glass bulk. When the defocus is down to −0.3 mm in Fig. 2(b), clear melting traces are observed at the metal side together with interfacial micro-cracks. Further increasing the defocus to −0.5 mm leads to the formation of coalesced pore defects within the glass matrix and an increased number of cracks, as visible in Fig. 2(c). With continued defocus increase to −0.7 mm in Fig. 2(d), the pore defects expand and distribute in a quasi-periodic pattern along the welding path and are accompanied by prominent interfacial cracking. Overall, the primary defect types evolve from fine cracks within the glass bulk to dominant interfacial pores and severe cracking, indicative of a shift in the underlying formation mechanism.

Theoretical calculations indicate that the peak power density at the focus under the experimental laser parameters significantly exceeds the nonlinear absorption threshold of fused silica. For a Gaussian beam, the peak power density at the focus, F_0, can be estimated by:

$$F_0 = \frac{2P}{\pi {\omega_0}^2 f_{rep} \tau}$$

where P is the average power, ω_0 is the beam radius at the focus as 20μm, f_{rep} is the repetition rate, and τ is the pulse width. Substitution yields F_0 on the order of 10^{13}W/cm^2, exceeding the nonlinear absorption threshold of fused silica as 10^{12}W/cm^2 approximately.

Furthermore, the range over which the laser maintains high power density within the medium is determined by the Rayleigh length Z_R, calculated as:

$$Z_R = \frac{\pi {\omega_0}^2}{\lambda}$$

Where λ is the laser wavelength. The calculated Z_R for this experiment is approximately 1.2mm. This implies that within a total range of about $2Z_R$ around the focal plane, the laser power density remains sufficient to induce nonlinear absorption in the glass. Therefore, under all experimental conditions, nonlinear absorption of the laser energy by the glass persistently occurs along the energy deposition channel traversing the glass, and its intensity is most severe near the focal spot.

The fundamental change in the macroscopic interfacial morphology with varying defocus is attributed to a shift in the dominant energy distribution mechanism. The total laser energy absorbed by the glass, E_{glass}, is the sum of energy deposited via linear and nonlinear absorption processes. However, for fused silica glass at a wavelength of 1030nm, the linear absorption is negligible. Therefore, E_{glass} is almost entirely governed by the nonlinear absorption process within the irradiated volume:

$$E_{glass} = \iiint_V \beta I^n \, dxdydz$$

where β is the nonlinear absorption coefficient, V represents the entire irradiated volume of glass, I is the spatial distribution of the laser intensity within the glass, and n is the order of the nonlinear absorption ($n \geq 2$, n=2 for two-photon absorption).

This formulation explains the observed decrease in E_{glass} as the focal plane deepens. Since the nonlinear absorption βI^n is intensely localized near the focus. As the focus shifts downward, this peak-intensity zone is compressed into a shorter segment of the beam path. Consequently, the beam interacts over a reduced length with the high-intensity region where absorption is most efficient. This shortens the effective absorption path, reducing the value of the volume integral and hence E_{glass}, despite an unchanged total geometric path.

Consequently, as the defocus distance decreases, the total laser energy absorbed by the glass shows a decreasing trend. This means that a greater portion of the laser energy penetrates the glass and acts on the metal interface. Simultaneously, the laser spot size on the metal surface changes very little within this defocus variation range. Therefore, the laser energy density incident on the metal surface increases significantly as the defocus distance decreases.

When this energy density, increasing with decreasing defocus, rises sufficiently to cause intense vaporization of the metal, it triggers a qualitative change in the dominant mechanism. The confined metal vapor, unable to expand freely due to the close coverage by the overlying glass, forms a high-pressure confined vapor. This high-pressure confined vapor exerts intense thermo-mechanical coupling effects on the glass interface, thereby dominating the reshaping of the interfacial morphology: on one hand, causing local melting of the glass to form micro melt pools; on the other hand, the associated thermal shock and residual stress lead to the generation of micro-cracks. Ultimately, features such as these micro melt pools, formed under this mechanism, become the characteristic morphology at large negative defocus distances.

3.2. Typical morphology at joint interface

To further study the metallurgical reactions at the interface, the glass was separated from the steel substrate by applying a controlled shear force. Specifically, the welded sample was mounted in a custom fixture that secured the quartz glass and the stainless steel block independently. A tensile testing machine was then used to apply a load, with the force vector aligned within the welding plane to induce pure shear separation. The morphology of the joint was observed using scanning electron microscopy. The main welding process parameters in the experiments are as follows: 1000 kHz pulse frequency, 16 W laser power, −0.5 mm defocus distance, and 50 mm/s welding speed. The morphology of the joint interface after removing the glass is showed in Fig. 3.

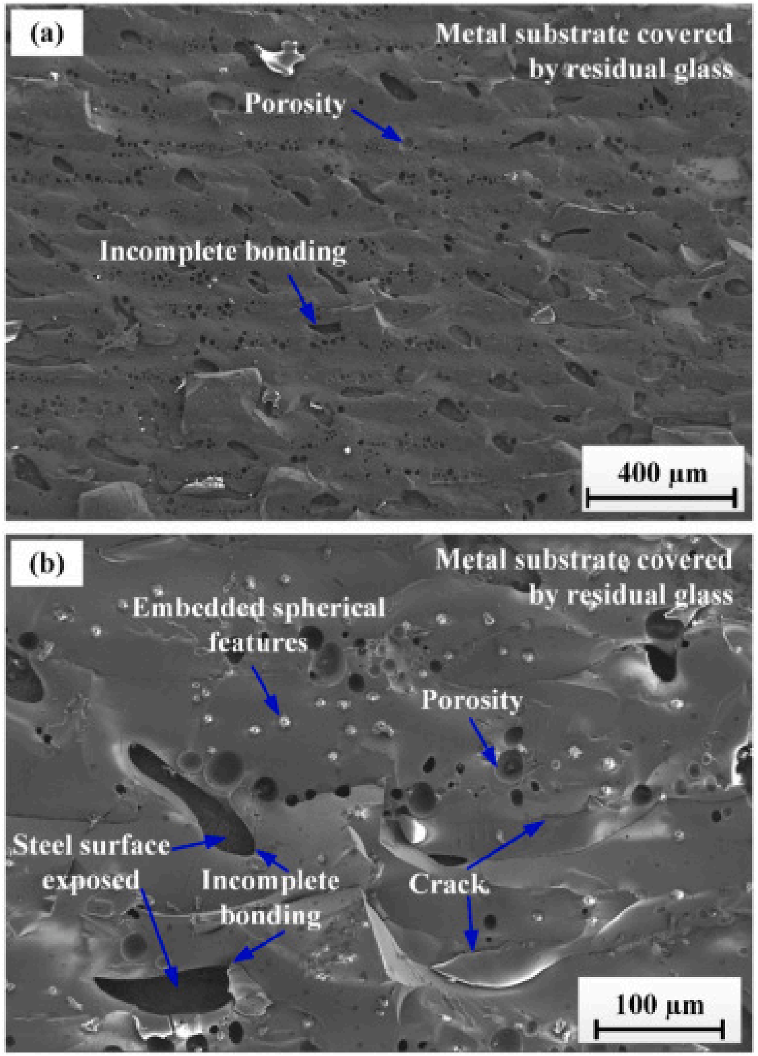

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Fig. 3. Morphology of the separated interface showing residual glass layer on the metal substrate: (a) at low magnification; (b) at high magnification.

As shown in Fig. 3(a), the typical defects in the interface layers consisted mainly of porosity and incomplete bonding defects. The shape of the porosity is mostly spherical in the interface reaction layers. These porosities are distributed in clusters within the glass. The formation of the porosity is closely associated with the interaction between laser and the metal at the interface.

As explained in Section 3.1, as the focal point moves downward, more laser energy is deposited at the metal surface, thereby increasing the laser energy density to the point where the metal undergoes intense vaporization. The high-pressure metal vapor generated within the molten metal zone is rapidly ejected. The tight confinement by the overlying glass forces this vapor to exert intense thermos-mechanical coupling on the glass interface. The vapor acts as a heat source, leading to localized melting of the glass. On the other hand, the vapor penetrates into the newly formed molten glass pool. Ultimately, during the rapid solidification of the weld, the vapor becomes entrapped within the glass volume, leading to the formation of the porosity. It can be also observed that an irregular plane is formed between the removed glass and stainless steel. No obvious reaction products were found in this area. From this, it can be inferred that this location should be an incomplete bonding defect. This bonding failure may also be attributed to the high-pressure metal vapor. The expanding vapor can exert sufficient force to displace the molten glass pool, thereby preventing localized contact and effective bonding with metal surface. From Fig. 3(b), it can be found that spherical granular products are embedded inside the glass. It might be metallic spatter particles resulting from the intense vaporization and ejection of molten stainless steel, which then intrude into the molten glass during the laser welding process. These particles are likely to act as crack initiation sites at the interface, as evidenced by the micro-cracks found encircling some of them. A detailed analysis will be presented in Section 3.5. Moreover, the cracks are clearly detected around the porosity. The formation of cracks inside the glass depended mainly on the thermal stress induced by the great thermal gradients inherent to the welding process. Critically, the stress is further amplified by direct mechanical loading from contained expanding metal vapor. The inherent high brittleness and low fracture toughness of quartz glass prevent the dissipation of this combined stress through plastic deformation. Consequently, the stress concentration finds relief through the propagation of micro-cracks, with the pore interfaces acting as natural initiation sites.

3.3. Microstructural evolution

In order to microstructural evolution of welding defects at the interface, the process experiments on pulse laser power were carried out on dissimilar materials of stainless steel and quartz glass. A single-pass welding strategy was employed to obtain a clearer picture of the fundamental laser-material interaction. The main welding process parameters in the experiments are as follows: 1000 kHz pulse frequency, −0.3 mm defocus distance, and 50 mm/s welding speed. The microstructure at the weld fracture with different pulse power levels is demonstrated in Fig. 4.

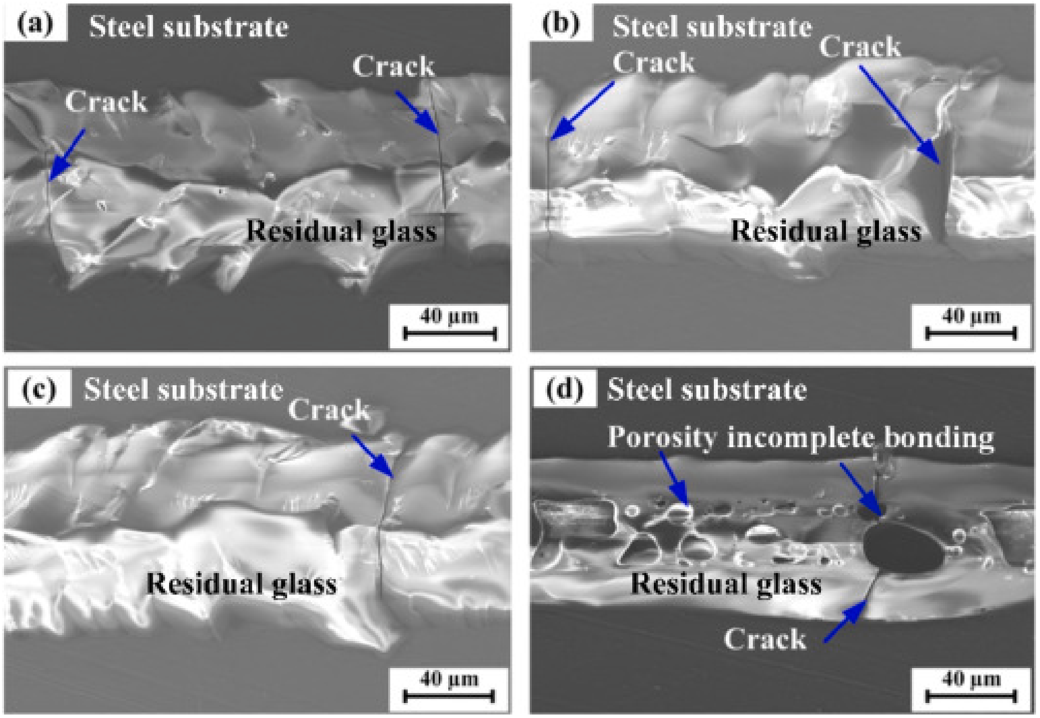

Fig. 4. Weld fracture at different pulse power levels: (a) 8W; (b) 12W; (c) 16W; (d) 20W.

As shown in Fig. 4(a), it can be clearly observed that the block glass is inlaid in the stainless steel matrix, which achieved a good combination between glass and metal. The embedding of the glass in the joints can serve as a binding function, enhancing the bonding force at the interface. It can be also found that the cracks occur inside the glass and spread along the direction perpendicular to the weld bead. It may be closely associated with the residual stress inside the glass. It may be closely associated with the residual stress from the thermal cycle inside the glass. The fact that only these longitudinal cracks are visible in this post-failure interface suggests that cracks in other orientations likely already contributed to the joint's failure under external load and are not present in this fracture surface. After laser radiation, the glass begins to melt and solidify under the influence of metal vapor. Because the glass is highly brittle, the thermal stress generated during the glass solidification process is failed to be completely eliminated, easily inducing the formation of crack sources at the defect location. As the residual stress increases, micro-cracks begin to proliferate and spread rapidly, finally forming large-sized cracks.

As the pulse laser power increases, the volume of residual glass and the contact area between it and the metal surface slightly increased, as observed in Fig. 4(b) and (c). At an appropriate defocus distance, such as −0.3 mm, a certain degree of laser power enhancement promotes the melting of the glass volume near the interface, resulting in forming a broader effective bonding area. No significant variation in the type, quantity, or distribution of defects is noted. This indicates that when energy deposition primarily occurs within the glass, increasing power mainly augments the total energy absorbed by the glass without fundamentally altering the interfacial formation mechanism. However, when pulsed laser power increases to 20 W, a large number of porosities and cracks are formed at the joint surface in Fig. 4(d). The crack defect originates from the porosity boundary and spreads towards the periphery. This transition suggests a shift in the dominant energy deposition mechanism: the higher power elevates the energy density at the metal surface beyond the threshold for intense vaporization. The resulting confined metal vapor governs the interfacial morphology. During this process, the vapor penetrates into the molten glass, and portions become trapped as solidification proceeds too rapidly for complete escape, leading to porosity. The existence of porosity defects leads to a reduction in the bonding force at the interface. Under the action of residual stress, cracks will form at the edge of the porosity and rapidly spread to the periphery.

3.4. Energy spectrum analysis

The occurrence of the observed welding defects is closely associated with the dynamic behavior of confined metal vapor and its thermomechanical interaction with the glass interface, which is further reflected in the elemental distribution characteristics around the defects. In order to further explore the causes of the formation of defects, energy spectrum analysis is used to analyze the characteristics of elemental changes around defects. EDS analysis was performed on cross-sectional specimens, comparing the elemental composition between the inner wall of exposed pores and the bulk glass within the weld seam (Fig. 5).

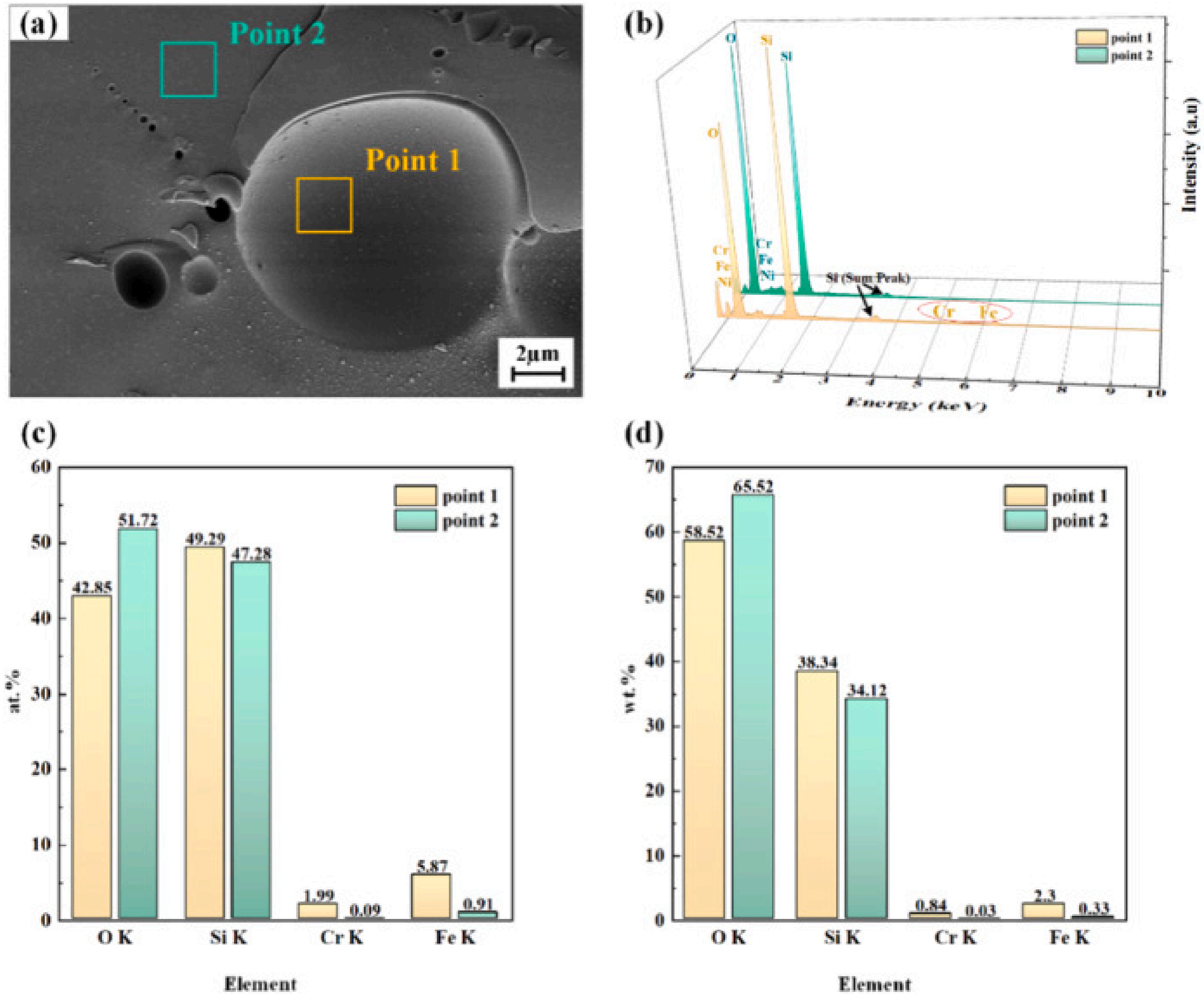

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Fig. 5. EDS results at different positions on the glass fracture surface: (a) positions for EDS point analysis; (b) comparison of EDS spectra at the two positions; (c) elemental atomic ratio; (d) elemental weight ratio.

To investigate the elemental interdiffusion and metallurgical behavior at the welding interface, characteristic elements representing the base materials were selected for EDS analysis: O and Si from the quartz glass side, along with Fe and Cr from the 304 stainless-steel. Their mass and atomic ratios were determined to characterize the interfacial reactions.

As shown in Fig. 5(b), significant levels of Fe and Cr from the stainless steel are detected on the pore inner wall at **Point 1**, which cannot be explained by measurement error. The quantitative elemental ratios presented in Fig. 5 (c) and **(d)** confirm this composition contrast. It is mainly attributed to the dynamic behavior of confined metal vapor at the interface. During the welding process, the metal vapor involving Fe and Cr can be generated due to the interaction between laser and stainless steel. After the metal vapor/are ejected from the molten metal zone, it acts on the glass, thus inducing its melting. The vapor subsequently penetrates the glass melt, driven by its high-pressure state at the interface. During rapid solidification, Fe and Cr atoms condense and are retained as solid deposits on the pore walls. However, the volume initially displaced by the expanding vapor fails to be fully replenished by the surrounding glass melt due to its limited fluidity under fast cooling conditions, resulting in the formation of porosity structure.

It can be found in Fig. 5(c) and **(d)** that the chemical elements near the porosity (**Point 2)** are predominantly consistent with that of the quartz glass base material. A comparative analysis reveals that the contents of chemical elements in **Point 2** are relatively close to that of the glass matrix. A small amount of Cr and Fe elements are also detected near the porosity. It might be caused by the migration of elements during the welding process.

The presence of porosity significantly alters the stress distribution within the joint. During welding, heat accumulation combined with the thermal energy delivered by metal vapor into the glass melt generates substantial stress. During solidification, the rate of stress generation from constrained thermal contraction exceeds the material's capacity for stress relaxation through viscous flow, the stress concentrates at pore edges and is released through the formation of micro-cracks. This process exhibits directionality correlated with the solidification front, resulting in micro-cracks around pores showing consistent orientation, as evidenced in Fig. 6(a). Under the influence of residual stress, these micro-cracks may propagate further to connect with adjacent pores or other cracks, as shown in Fig. 6(b), potentially evolving into macroscopic cracks that traverse the weld seam and severely compromise the joint's mechanical performance.

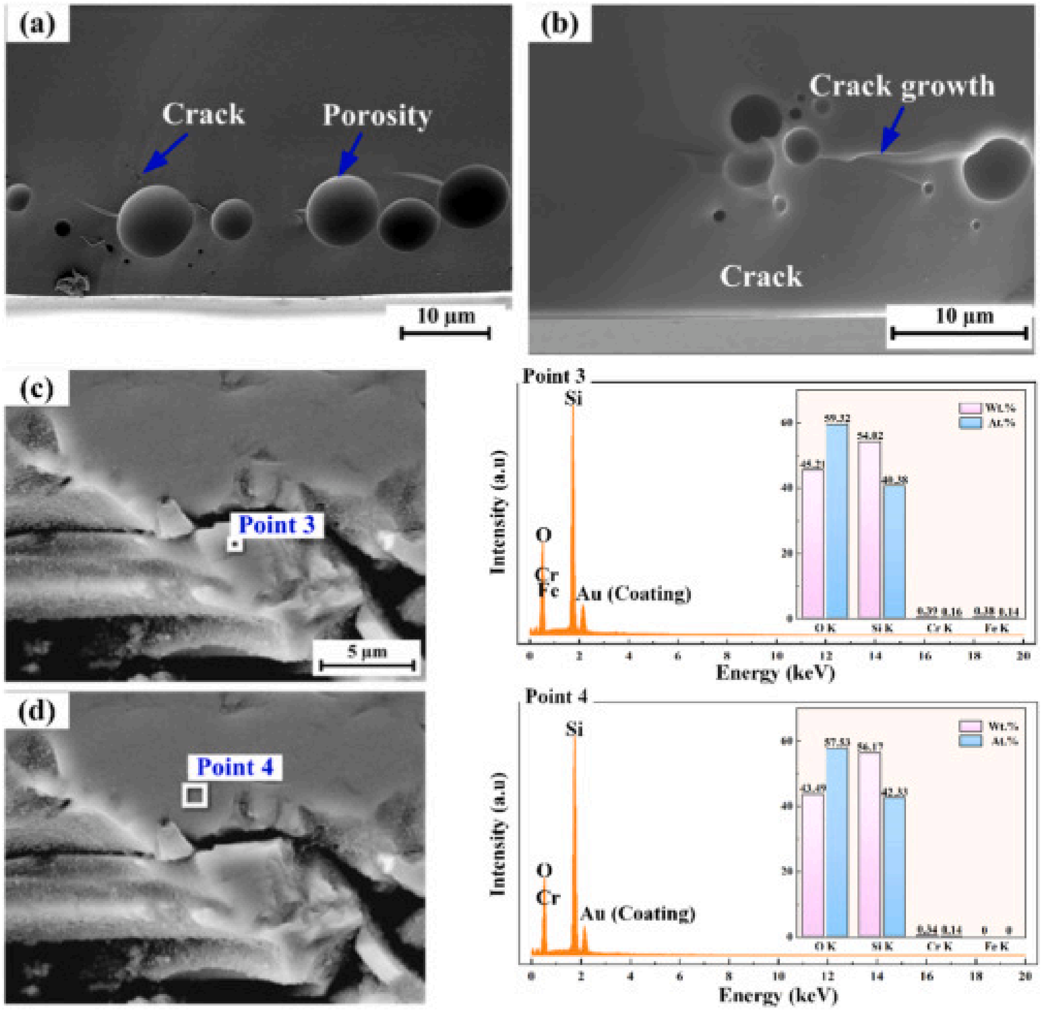

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Fig. 6. Cracks induced by microstructural stresses at the interface: (a) micro-cracks; (b) crack growth; (c) and (d) quantitative EDS results near the cracks.

Elemental analysis via EDS of both sides of these cracks reveals compositions with Si:O atomic ratios approximating 2:3, as shown in Fig. 6(c) and (d), where the Au peaks originate from the conductive gold coating. This result indicates a deviation from the 1:2 ratio of base quartz glass. This consistent non-stoichiometry suggests the formation of oxygen-deficient glass structures under the extreme thermal conditions. Based on the EDS findings, X-ray photoelectron spectroscopy (XPS) analysis was conducted to further investigate the chemical states of the elements at the fracture interface, as presented in Fig. 7.

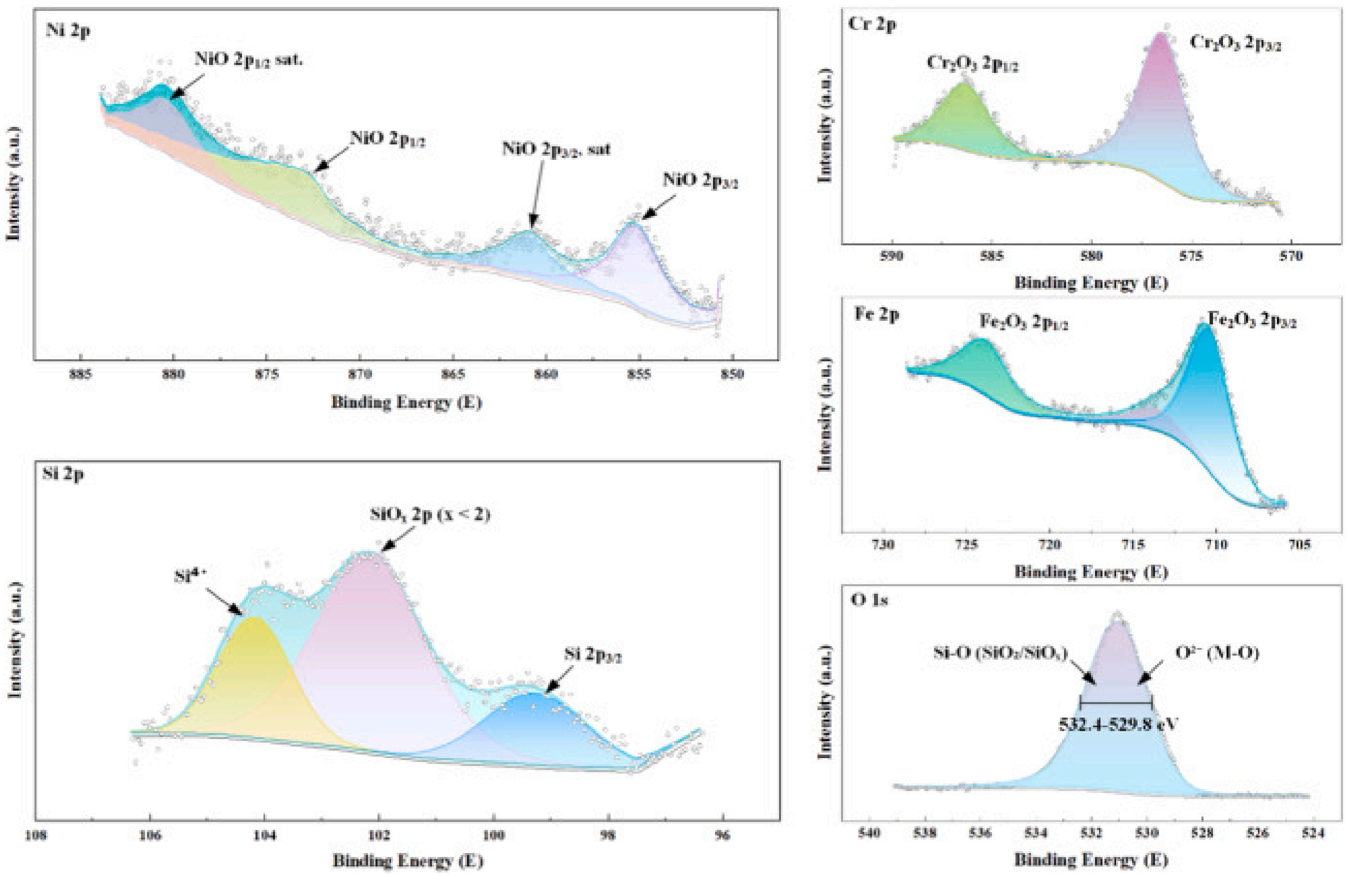

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Fig. 7. XPS results at the joint interface.

The XPS analysis confirms the presence of metal oxide bonding energies in the spectra of stainless-steel elements. The Ni 2p region shows the main NiO $2p_{3/2}$ peak at 855.0eV. In the Cr 2p spectrum, the Cr_2O_3 $2p_{3/2}$ peak appears at 576.8eV. The Fe 2p region exhibits the Fe_2O_3 $2p_{3/2}$ peak at 711.0eV. In the Si 2p region, a distinct peak between the Si^{4+} and Si $2p_3/_2$ peaks, together with the non-stoichiometric Si:O atomic ratio obtained by EDS, indicates the formation of an oxygen-deficient structure, denoted as SiO_x. The O 1s spectrum exhibits a broad peak with a FWHM spanning 532.4-529.8 eV, encompassing both the lower binding energy M−O bonds from metallic oxides and the Si-O bonds from both SiO_2 and SiO_x. These XPS results provide definitive evidence for the reduction of fused silica under extreme thermal conditions, corroborating the oxygen deficiency suggested by EDS analysis and confirming the formation of metastable, oxygen-deficient glass structures at the crack interface.

3.5. TEM analysis

In order to further investigate the interfacial microstructure in laser melting zones, an electron-transparent cross-section sample of the 304 stainless steel-quartz glass interface was prepared using focused ion beam (referred to as FIB) at the location marked by the yellow rectangle in Fig. 8(a). TEM observations of this specimen reveal the presence of dispersed particulate phases within the glass matrix adjacent to the interface, as shown in Fig. 8(a) and (b). These particles, ranging from several nanometers to nearly one hundred nanometers in diameter, are distributed in the glass side near the boundary region.

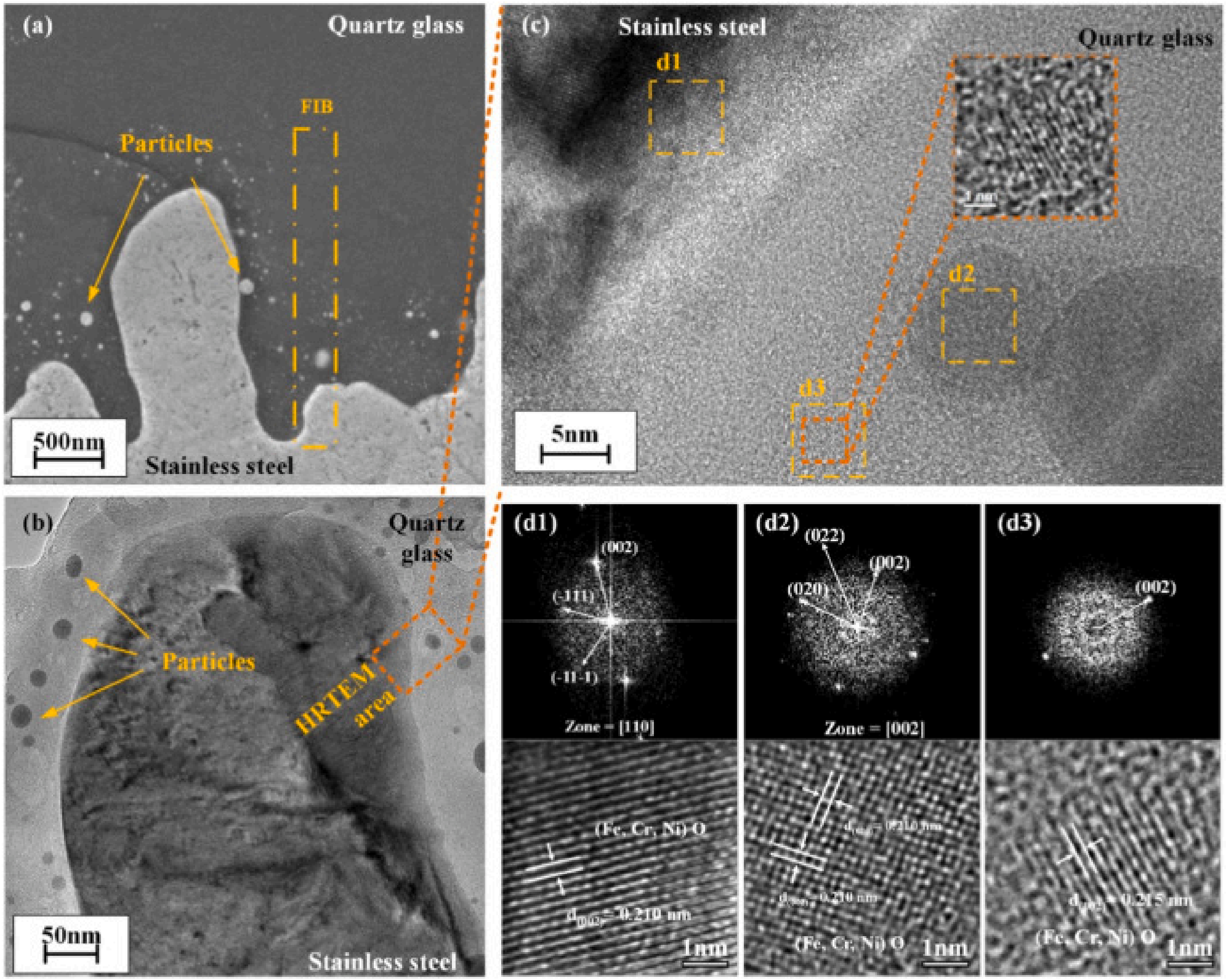

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Fig. 8. TEM analysis at the stainless steel/quartz glass interface: (a) cross-sectional SEM of the interface with FIB location; (b) TEM of the interface with high resolution TEM (HRTEM) region; (c) HRTEM with three crystalline areas marked; (d) Crystallographic analysis of marked regions.

HRTEM observation was conducted on the orange-red region marked in Fig. 8(b), with the results presented in Fig. 8(c). Three structural features were identified: a bulk crystalline phase on the stainless-steel side of the boundary, spherical crystalline particles embedded within the amorphous silica matrix, and a finer particulate phase with indistinct crystallographic features, presumably precipitated from the amorphous matrix. The finer particles in the marked region d3, which are smaller than 3 nm in diameter, exhibit a morphology and distribution similar to the larger spherical ones. This similarity suggests they share the same phase origin, despite their weak diffraction signature shown in Fig. 8(d3).

TEM-based phase analysis was systematically performed to identify the characteristics of these structures, with the results presented in Fig. 8(d). The diffraction patterns from all crystalline regions match those of NiO, with measured lattice spacings in close agreement. The slight expansion in lattice parameters (approximately 0.02-0.03 Å) could be attributed to the formation of a (Fe, Cr, Ni) O solid solution, where Fe^{2+} and Cr^{3+} ions may have substituted for Ni^{2+} in the lattice. The predominance of the NiO-based structure is likely due to kinetic advantages under the extreme non-equilibrium solidification conditions. Specifically, the simple rock-salt structure of NiO may have enabled faster nucleation and crystal growth compared to more complex oxide phases. This rapid crystallization would have allowed NiO to form the initial crystal framework, into which the more abundant Fe^{2+} and Cr^{3+} ions subsequently incorporated, resulting in the observed solid solution with its expanded lattice parameters.

High-angle annular dark-field (HAADF) imaging was employed to obtain images with high compositional contrast, followed by EDS mapping for elemental analysis as shown in Fig. 9. Based on the TEM and EDS mapping results, these dispersed particles are confirmed to contain stainless steel elements (Fe, Cr, Ni), indicating their metallic origin. The elemental mapping reveals that the distribution of oxygen is more extensive compared to silicon and shows substantial overlap with the metallic elements (Fe, Cr, Ni). This suggests the formation of complex oxide phases, likely a (Fe, Cr, Ni) O solid solution as identified previously, at the expense of oxygen from the surrounding silica glass matrix. This observation provides further explanation for the oxygen deficiency previously identified by EDS analysis at the crack regions in the preceding section, supporting the hypothesis of oxygen consumption during interfacial reactions.

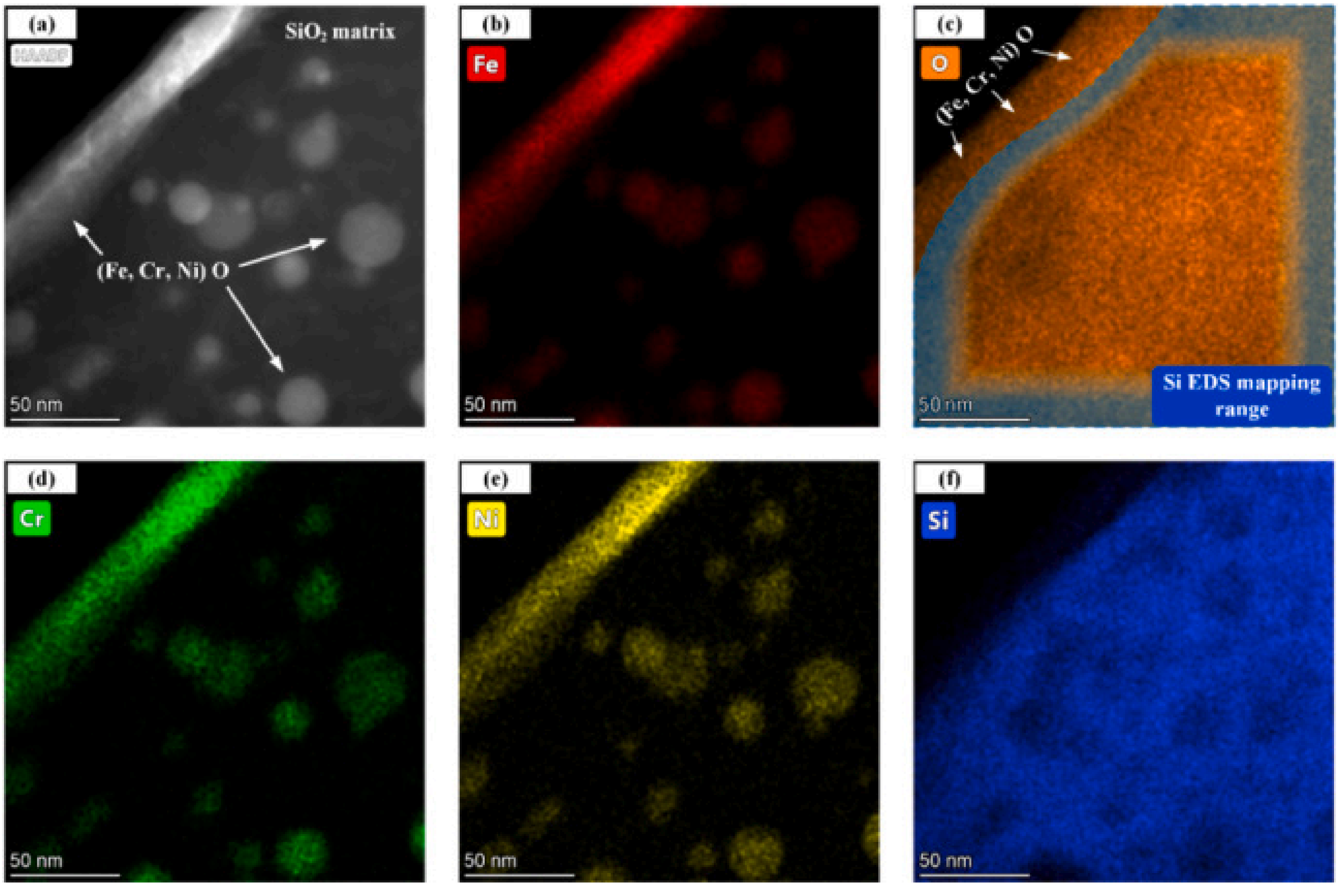

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Fig. 9. Nanoscale elemental distribution across the interface by STEM-EDS mapping: (a) HAADF image of the interface and EDS mapping of (b) Fe, (c) O, (d) Cr, (e) Ni, (f) Si.

Based on TEM and EDS analyses, the presence of crystalline (Fe, Cr, Ni) O nanoparticles within the silica glass matrix is likely detrimental to the joint's mechanical integrity. The primary concern arises from the significant thermophysical property mismatch between the particles and the glass, which induces localized stress concentrations around the particles during welding or under load. These stresses can promote micro-crack initiation, as observed around particles in the interfacial region as shown in Fig. 3(b), providing preferential paths for crack propagation and potentially leading to premature failure. Furthermore, the formation of these oxide particles consumes oxygen from the silica network. The resulting oxygen-deficient SiO_x structure is inherently less stable than stoichiometric SiO_2. The oxygen deficiency leads to a defective atomic network characterized by broken bonds and coordination defects, which collectively render the local matrix more susceptible to fracture. Therefore, these interfacial nanoparticles, which act as both sites of localized stress concentration and contributors to local matrix embrittlement, constitute a microstructural factor that undermines the mechanical strength and reliability of the weld.

3.6. Defect formation mechanism

As indicated by the experimental observations and characterization results above, the ultrafast laser welding of quartz glass to 304 stainless steel produces a series of interfacial morphologies and defect structures fundamentally governed by the spatial competition between two primary energy deposition mechanisms, as conceptually summarized in Fig. 10. The transition between these competing mechanisms can be modulated by the defocus distance and laser power parameters, where their variation drives a systematic shift in the dominant energy deposition regime from nonlinear absorption within the glass bulk to metal vaporization at the interface. These regimes are conceptually categorized as glass-volume-dominated, glass-metal interfacial, and metal-surface-dominated energy deposition.

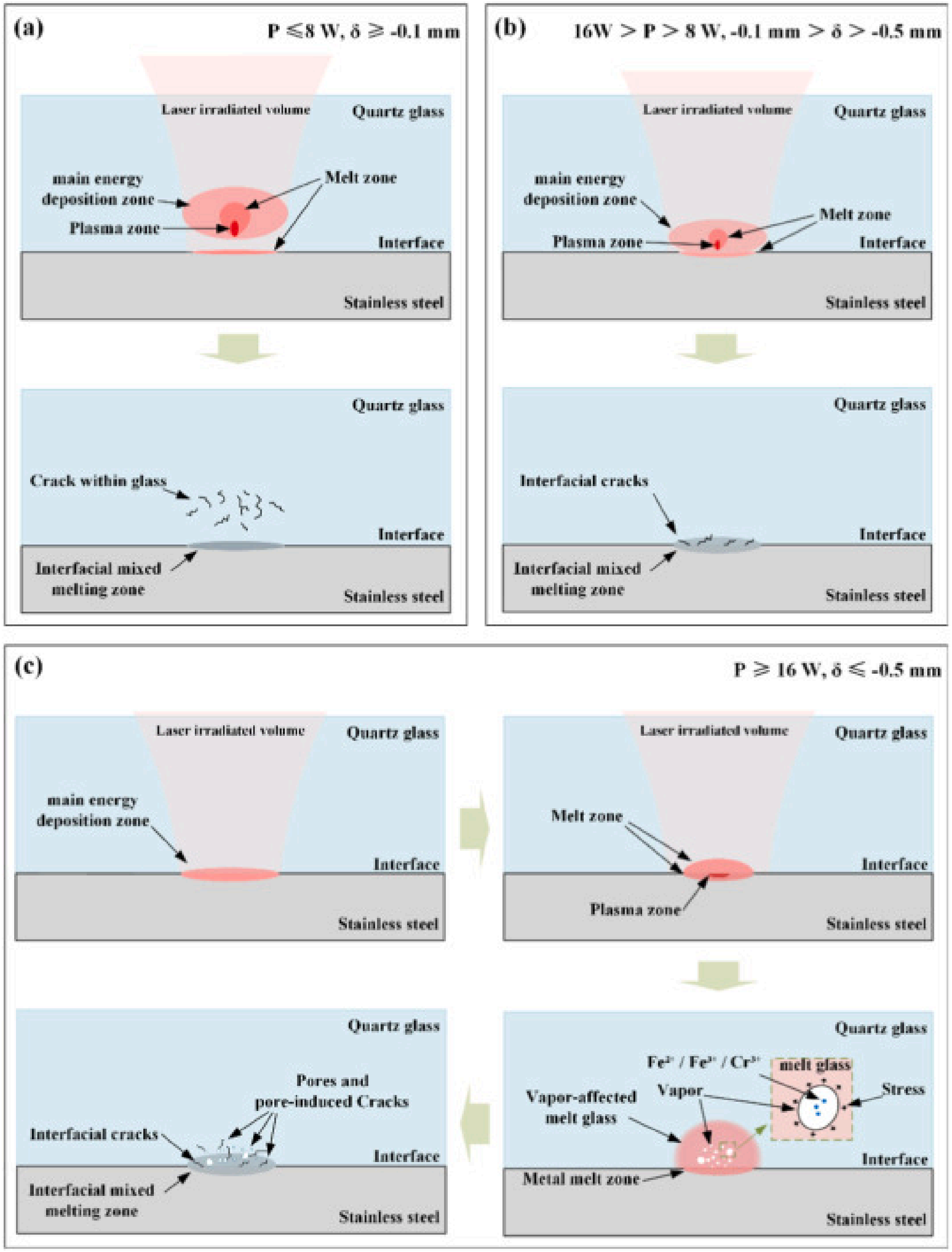

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Fig. 10. Schematic diagram of defect formation mechanism: (a) defect formation under glass-volume-dominated energy deposition; (b) defect formation under glass-metal interfacial energy deposition; (c) defect formation under metal-surface-dominated energy deposition.

To explain the transformation of the energy deposition mechanism within quartz glass, the nonlinear Kerr self-focusing effect at ultra-high peak power must be taken into account. This effect fundamentally determines the deposition behavior of energy by altering the actual focus position of the light beam. Based on the Gaussian beam formula provided in Section 3.1, the peak energy density at the focal plane and the metal-glass interface is determined. Besides, the Marburg-type formula is also adopted to calculate the equivalent focus offset distance (z_{nl}) caused by nonlinear propagation [40]:

$$z_{nl} = \frac{0.367 k_0 {a_0}^2}{\sqrt{\left(\sqrt{\frac{P_{peak}}{P_{cr}}} - 0.852\right)^2 - 0.0219}}$$

Among them, $k_0=2\pi n/\lambda$ is the wave number, a_0 is the radius of the beam on the incident surface of the quartz glass, calculated according to the Gaussian beam propagation formula, P_{peak} is the peak power of the laser pulse, and $P_{cr} \approx 3.0\times10^6$W is the self-focusing critical power of the quartz glass, obtained based on the standard data of pure quartz glass materials.

However, the standard Marburger formula is applicable to single-pulse conditions. In the experiment, the heat accumulation effect caused by a repetition frequency as high as 1000 kHz, as well as the heat conduction and seed electrons provided by the metal below, all enhance the nonlinear effect. Therefore, Two correction factors greater than unity, C_1 (accounting for the thermal accumulation effect due to repetition rate) and C_2 (representing the influence of the metal presence), are incorporated into the formula. In the formula, that is, the corrected offset distance is $z'_{nl}=z_{nl}\times C_1\times C_2$. With these factors, the focal shift Δz was derived via the compound lens focal formula [40].

$$\Delta z = f^* - \frac{1}{\frac{1}{f^*} + \frac{1}{z_{nl}}}$$

Here, f^* represents the linear propagation distance within the quartz glass.

For instance, under conditions such as a −0.1 mm defocus and 16 W power, nonlinear effects initiate an upstream focal shift of at least ~117 μm within the glass. This shift leads to enhanced absorption and energy deposition above the intended focal plane, driven by nonlinear ionization and incipient filamentation. The process culminates in the laser self-focusing into a filament that continues to propagate upstream. When the laser focus is within the glass, including slightly below the interface, the resulting glass-volume-dominated energy deposition regime is characterized by nonlinear absorption within the silica matrix itself as shown in Fig. 10(a). The primary defect in this regime manifests as interlaced micro-cracking networks originating within the glass bulk, a direct consequence of the intense localized thermal stress that cannot be relieved through the material's limited viscous flow during rapid solidification.

The transition between regimes is also governed by the material ablation thresholds, which determine when metal vaporization begins to dominate. The commonly used D^2 method is adopted to measure the damage threshold of quartz glass and 304 stainless steel under the action of 1064 nm ps laser. The square of the damage diameter D^2 of the laser-ablated material and the laser energy density E satisfy the following relationship [41]:

$$D^2 = 2\pi {\omega_0}^2 \ ln\left(\frac{E}{E_{th}}\right)$$

Here, E_{th} represents the energy density threshold of laser ablation. It is calculated that the ablation threshold of quartz glass is 0.511 J/cm^2, lower than the typical range (~1-2 J/cm^2), attributable to the thermal accumulation effect brought by the 1000 kHz repetition frequency. The ablation threshold of 304 stainless steel is 0.3 J/cm^2, slightly below the general value (~0.5 J/cm^2), due to the thermal accumulation effect of multiple pulses and the enhanced interference caused by the Fabry-Perot cavity between the bi-material polished interfaces. It is crucial to emphasize that the onset of metal ablation does not signify an immediate shift to a metal-surface-dominated deposition mode. Instead, it marks the initial participation of metal vapor in joint formation. As the laser power increases or the focal point moves, which increases the laser power density irradiated on the metal surface, the pressure generated by this metal vapor intensifies progressively, eventually becoming the primary driver of defect formation.

Based on the interplay of focal shift and ablation thresholds, the three energy deposition regimes manifest under specific parameter combinations. As the focal plane shifts downward (for example, −0.3 mm, 16 W), a hybrid glass-metal interfacial energy deposition regime emerges. This regime features a balanced energy distribution between the glass bulk and the metal interface, as shown in Fig. 10(b). Under the fixed defocus condition of −0.3 mm (Section 3.3), increasing the laser power from 8 W to 20 W raised the calculated peak energy density on the stainless steel surface from ~0.28 J/cm^2 to ~0.70 J/cm^2. When the power remains below 16W, the resulting metal vapor pressure is insufficient to dominate defect formation, allowing the "glass-metal interface" mechanism to prevail. This condition enables moderate nonlinear absorption near the welding zone at the lower portion of the glass while allowing sufficient energy transfer to the interface to promote metallic bonding, yet maintains the vapor formation at a manageable level that mitigates the risk of excessive defect generation. This balanced energy allocation results in fewer and less severe defects compared to the extreme regimes, making it a promising parameter window for achieving reliable joints.

As the defocus increases beyond about −0.5 mm or the laser power exceeds 16W, the primary mechanism undergoes a transition to metal-surface-dominated energy deposition, where the energy deposition at the metal interface intensifies to the point where the vapor pressure becomes sufficient to govern joint formation and defect evolution, as shown in Fig. 10(c). This initiates a series of vapor-dominated processes: the instantaneous generation of metal vapor, which is physically constrained by the overlying glass, forms a high-pressure confined vapor. This vapor becomes the primary agent reshaping the interface, and its behavior directly dictates the defect morphology. A significant fraction of this vapor infiltrates the molten glass pool, and the ultrahigh cooling rates inherent to the process freeze it in place, leading to the formation of spherical vapor-entrapped porosity. This vapor infiltration and entrapment creates substantial thermomechanical coupling at the interface. The combined effect of direct mechanical loading from the expanding vapor and residual thermal stress from steep thermal gradients generates intense localized stress. This stress is released in the stress concentration zones formed at pore boundaries, initiating micro-cracks that propagate around the pores. Under sustained stress, these micro-cracks can extend to bridge adjacent pores or other defects, potentially coalescing into macroscopic cracks that severely degrade mechanical integrity. Furthermore, the high-pressure vapor can mechanically displace the molten glass from the metal surface in localized areas, preventing intimate contact and resulting in incomplete bonding defects.

Based on the mechanistic analysis above, Fig. 11 schematically summarizes the progressive evolution of defect formation, illustrating the causal relationships between energy deposition mechanisms, physical processes, and the resulting defect systems.

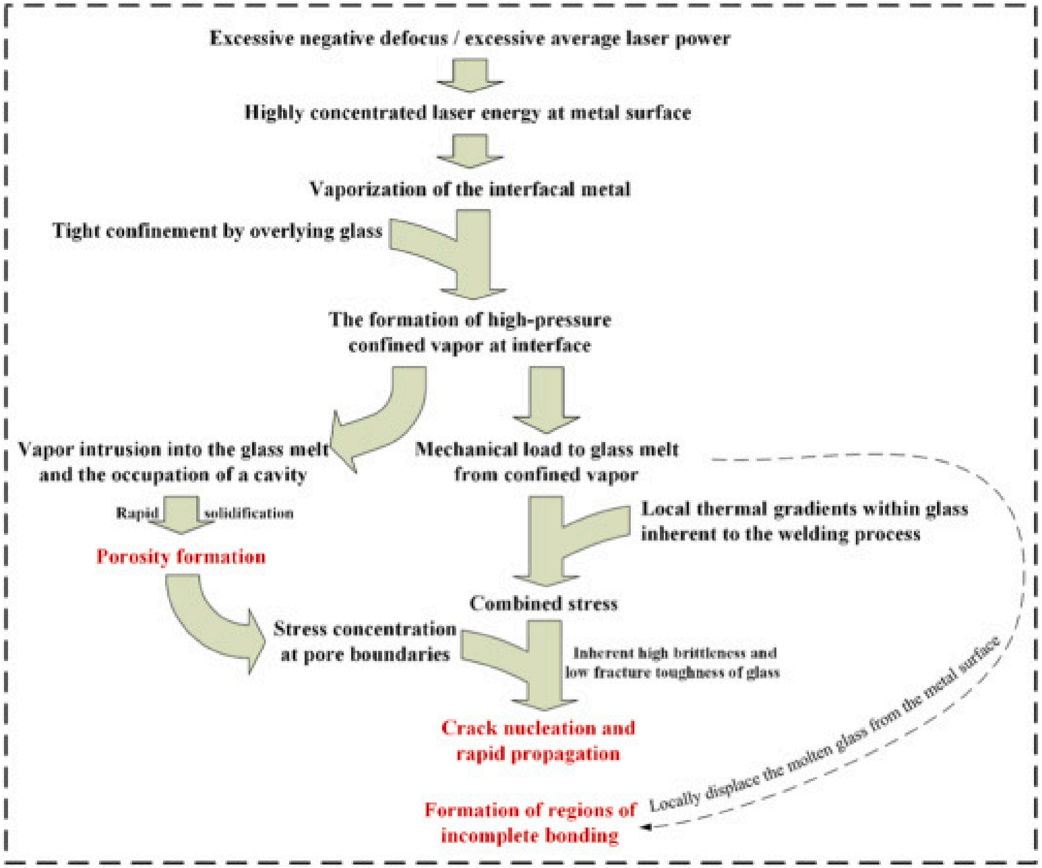

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Fig. 11. Process model of defect formation.

3.7. Possible improvements to the welding process

The correlation between weld morphology and performance, coupled with the established defect formation mechanism, provides a foundation for potential improvements to the welding process. The model confirms that the final weld quality is dictated by the competition between the vapor-driven mechanism and the bulk glass modification mechanism. This understanding directly informs the optimization strategy: to suppress the excessive vapor-driven mechanisms while promoting sufficient interfacial melting. Since defocus distance and laser power collectively govern the laser energy density incident on the surface of stainless steel. Essentially, these parameters influence defect formation by altering the spatial distribution of laser energy. Therefore, optimizing the spatial energy distribution is the key to enhancing the process. By tailoring the energy distribution, it becomes possible to ensure adequate energy delivery at the interface to promote melting and gap filling while maintaining metal vapor generation within a controlled range, thereby mitigating defect formation and enhancing joint integrity.

The principle above was implemented in our process optimization through the following approach. A galvanometer scanner with a 100 mm focal length lens was employed to achieve a more concentrated laser energy distribution. Based on previous experiments and preliminary tests with the new setup, a strategy employing relatively low laser power with an appropriate defocus distance was implemented to maintain the laser energy density on the stainless steel surface below the threshold for excessive vaporization. Other potentially influential paraments, including the repetition rate, scanning speed, clamping load, and scanning pattern, were kept consistent with the prior, non-optimized experiments to ensure a direct comparison. The complete set of process parameters for both conditions is summarized in Table 3.

Table 3. Process parameters of the optimized and non-optimized conditions.

Variables	**Units**	**non-optimized conditions**	**optimized conditions**
focal length	mm	170	100
Pulse laser power	W	8; 12; 16; 20	6; 8; 10; 12
Defocus distance	mm	−0.1; −0.3; −0.5; −0.7	−0.1; −0.2; −0.3; −0.4; −0.5
Pressing force	N	400	400
Repetition frequency	kHz	1000	1000
Welding speed	mm/s	50	50

After welding, shear tests were conducted using an electrodynamic static fatigue testing machine to evaluate and compare the mechanical performance of the welded joints. The results are presented in Fig. 12. Using the new experimental setup and optimized welding process with parameters of 8 W laser power and −0.3 mm defocus distance, the best shear performance within the test series was achieved.For comparison, the unoptimized group achieved its highest strength using parameters of 12 W laser power and −0.3 mm defocus distance. The load-displacement curves from three repeated tests for each of these best-performing parameter sets are compared in Fig. 12(a). As shown in Fig. 12, the average maximum load reached 332.68 N (approximately 534% of the unoptimized group's value of 62.25 N), with the peak load reaching 378.41 N (approximately 517% of the previous maximum of 73.21 N). Other process parameters were as follows: 1000 kHz pulse frequency, 50 mm/s scanning speed, and 400 N clamping load.

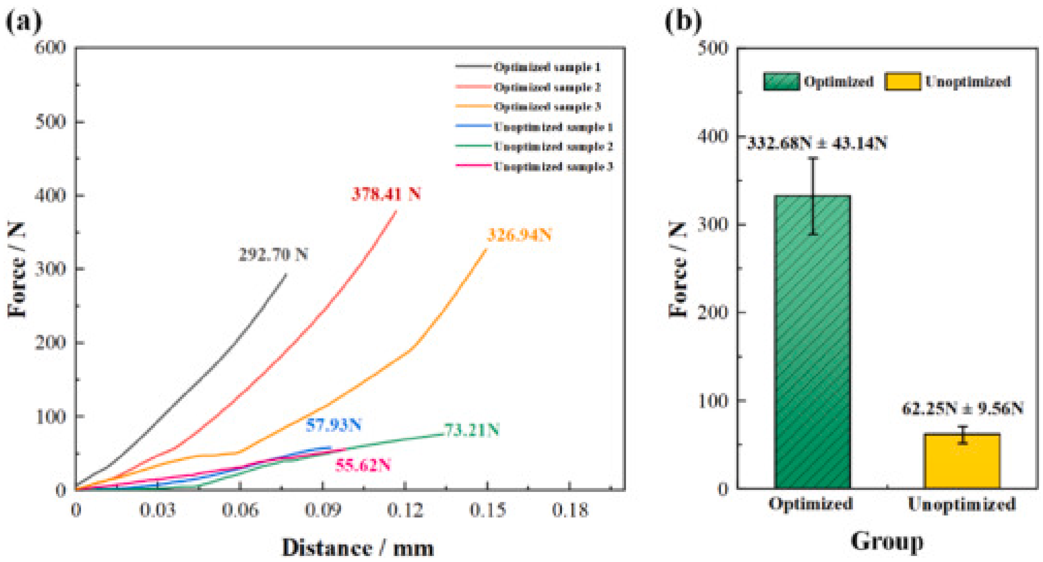

Fig. 12. Shear performance of best parameters: (a) load-displacement curves for three identical tests from optimized and unoptimized groups; (b) comparison of the mean maximum load.

The load-displacement curves in Fig. 12(a) exhibit distinct failure characteristics. The unoptimized samples show highly consistent curves, failing at low loads with little variation. This suggests that failure was dominantly governed by the widespread pre-existing defects (pores and cracks), leading to predictable, defect-driven fracture. In contrast, the optimized samples, while achieving higher failure loads, display greater variation in their failure behavior. This increased scatter is attributed to the competition between the inherently weak plane (the glass-metal dissimilar material interface) and the sporadic, residual micro-defects. The stochastic nature of these residual micro-defects can be attributed to several micro-scale sources: (1) localized contamination or oxide spots on the original metal surface, and (2) slight inhomogeneities in the initial optical contact between the glass and metal. The latter originates mainly from residual microscopic peaks and valleys that persist after polishing. Stylus profilometry quantified this surface condition, showing a peak sharpness Rku of 4.9163 for the metal and 3.9790 for the glass. These local, sporadic non-uniformities, together with inherent specimen-to-specimen variations, are ultimately responsible for the relatively large scatter in the mechanical test results. Because the major defects are suppressed, the failure initiation site becomes stochastic, governed by the distribution of these subtle imperfections. This randomness manifests as the observed diversity in the load-displacement curves and is further corroborated by the relatively large error bars in the maximum load data for the optimized samples in Fig. 12(b).

To further investigate joint failure modes, the fracture morphologies of both joint types were examined, the macro and micro morphologies of the fracture surfaces are shown in Fig. 13. Fig. 13(a) and (b) present the macroscopic fracture morphologies of the joints after shear testing. The unoptimized joint in Fig. 13(a) exhibits an irregular interface with localized bright zones on the steel side and dark deposits on the glass side, where the bright reflections from the metal side suggest the presence of large-area residual glass phase, while the dark deposits on the glass side indicate substantial metal ablation during welding. The optimized joint in Fig. 13(b) shows a more uniform interface appearance. To gain deeper insight into the fracture mechanisms, SEM was performed on the characteristic regions of these fractures. As shown in Fig. 13(c), the fracture surface of the unoptimized sample features several pores, with characteristic radiating crack patterns emanating from their boundaries. This suggests that failure initiated and propagated preferentially from these pre-existing defects.

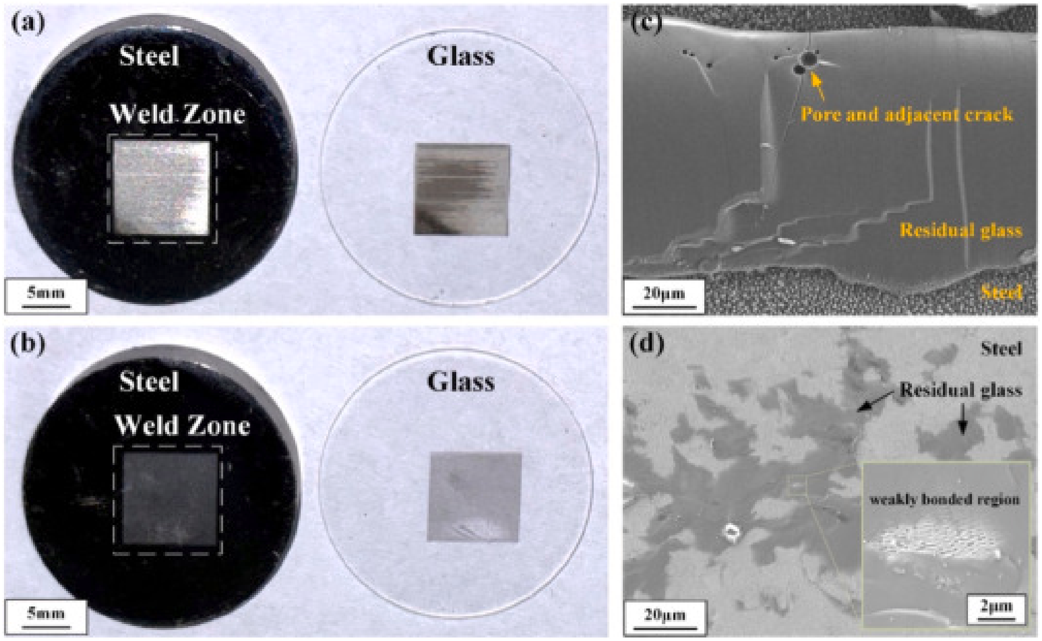

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Fig. 13. Comparative analysis of fracture surface morphologies: macroscopic view of fracture surface from (a) unoptimized sample and (b) optimized sample; SEM micrograph of fracture surface from (c) unoptimized sample and (d) optimized sample.

In contrast, as is shown in Fig. 13(d), the optimized samples show smoother fracture areas, the failure occurred mainly along the glass-metal interface. However, sporadic, localized weakly bonded regions likewise acted as initiation points, causing the crack to propagate either along the interface or into the glass matrix near the interface. The observed morphological features provide strong support for the proposed failure mechanism, indicating a clear transition in fracture behavior. While unoptimized joints failed through predictable, defect-dominated fracture, the optimized joints displayed failure propagation that varied between the bonded interface and sporadic residual defects.

Based on the comprehensive analysis of mechanical performance and fracture characteristics, the effectiveness of the proposed process optimization strategy is demonstrated. The corresponding improvement in joint is confirmed by the five-fold increase in average shear strength. This work presents a viable approach for enhancing ultrafast laser welding of stainless steel–quartz glass dissimilar materials through targeted control of energy deposition characteristics.

4. Conclusions

In this study, the infrared picosecond laser welding was conducted on dissimilar materials of quartz glass and stainless steel. The effects of welding parameters on weld formation were carefully investigated. The evolution law and forming mechanism of typical defects in glass-stainless steel welding joint was deeply explored. Moreover, possible improvements to the welding process have been proposed to achieve a significant improvement in the mechanical performance of the joint. The main conclusions are summarized as follows.

(1) The defocus distance and laser power govern defect formation by dictating the spatial distribution of laser energy. Specifically, the dominant mechanism shifts between the glass bulk and the metal interface, thereby determining the type and severity of defects.

(2) Porosity within the joint was primarily formed by the confined high-pressure metal vapor, generated under intense laser energy at the surface of metal, which penetrated into the molten glass pool and became trapped during rapid solidification.

(3) Micro-cracks initiated at the boundaries of pores due to stress concentration, and subsequently propagated under the combined action of residual thermal stress and direct mechanical loading from the expanding vapor, ultimately coalescing into macroscopic cracks.

(4) An "energy deposition mechanism competition" model was established, categorizing the process into three distinct regimes (glass-volume-dominated, glass-metal interfacial, and metal-surface-dominated), which successfully elucidated the physical origin of the various defect systems.

(5) Under the metal-surface-dominated energy deposition regime, stainless steel elements (Fe, Cr, Ni) entered the glass melt in the form of vapor and reacted with oxygen from the glass, forming (Fe, Cr, Ni) O complex oxide nanoparticles with NiO as the crystalline framework. This reaction consumed oxygen from the glass

matrix, leading to the formation of an oxygen-deficient SiO_x structure in the interfacial region, which was further confirmed by XPS analysis.

(6) Guided by this model, the welding process was optimized by tailoring the spatial energy distribution, which effectively suppressed defect formation and achieved a significant, over five-fold increase in the average shear strength of the joint.

CRediT authorship contribution statement

Shuye Zheng: Writing – original draft, Methodology, Investigation. **Chunming Wang:** Writing – review & editing, Supervision, Conceptualization. **Youting Gan:** Methodology, Investigation, Formal analysis. **Fei Yan:** Writing – review & editing, Validation, Funding acquisition. **Yu Huang:** Writing – review & editing, Resources, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people of organizations that can inappropriately influence our work.

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